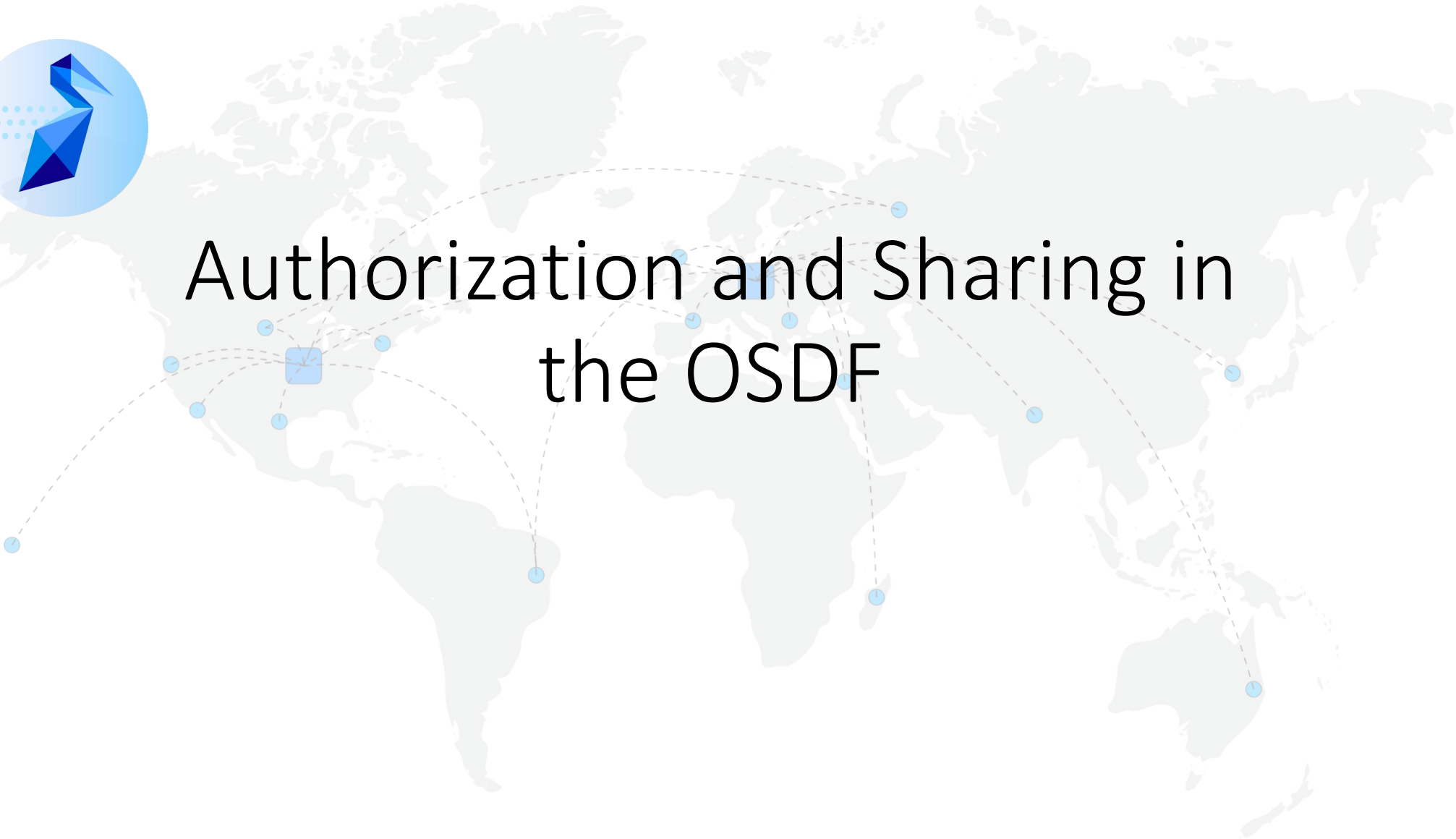




Authorization and Sharing in the OSDF





A national object delivery network

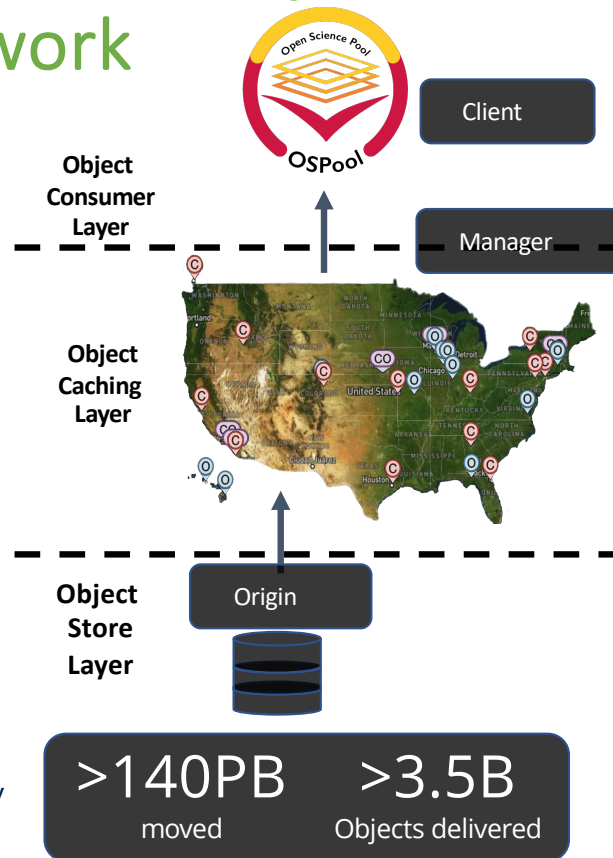


Who can benefit?

- **Open** to any federally-funded science repository or object store.
- **Researchers** can access/connect their data to local or national compute by browser or notebook
- **Data providers** can share via FAIR principles.
- **Compute providers** can cache datasets on-site
- **Community platform providers** can build gateways and portals to large-scale datasets.

Features

- No direct cost for integration.
- Supports automation of data-intensive workloads
- Unified name space preserves local autonomy
- Unified authorization infrastructure for restricting object access.



34 caches throughout the world, at points of presence within major compute centers and the global R&E networks (ESNet, Internet2).

Example use cases

- Enable large-scale workloads that have significant **reuse and redelivery** of objects at remote autonomous object stores.
- Manage loads on object stores and **avoid egress costs** via caching & staging.
- **DOE**: FNAL (LHC, DUNE), JLab, Fusion
- **NOAA**: Via AWS OpenData
- **NSF**: NCAR, NRAO, LIGO, EHT

Key concepts

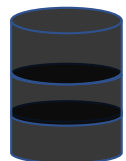
- **Object**: both data & non-data objects (e.g., containers, models, libraries).
- **Object store**: a filesystem, S3-compatible endpoint, or HTTP webserver.
- **Origin service**: connects the backend object store with national infrastructure. Objects are read (GET) and written (PUT) through Origin.

Object Store Layer

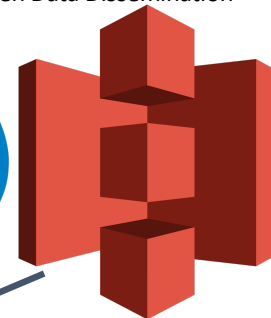


OSDF - Connecting The Nation's Data

NOAA Open Data Dissemination



The origin service connects storage to the OSDF.



Object Caching Layer

Data is cached for reuse, reducing the need for multiple egresses. Caches are embedded in the network.

The manager services coordinate the data access

Object Consumer Layer

Integrates well with national compute capacity, Jupyter notebooks, or a browser

Using a single toolset, researchers can fuse data from across multiple sources.



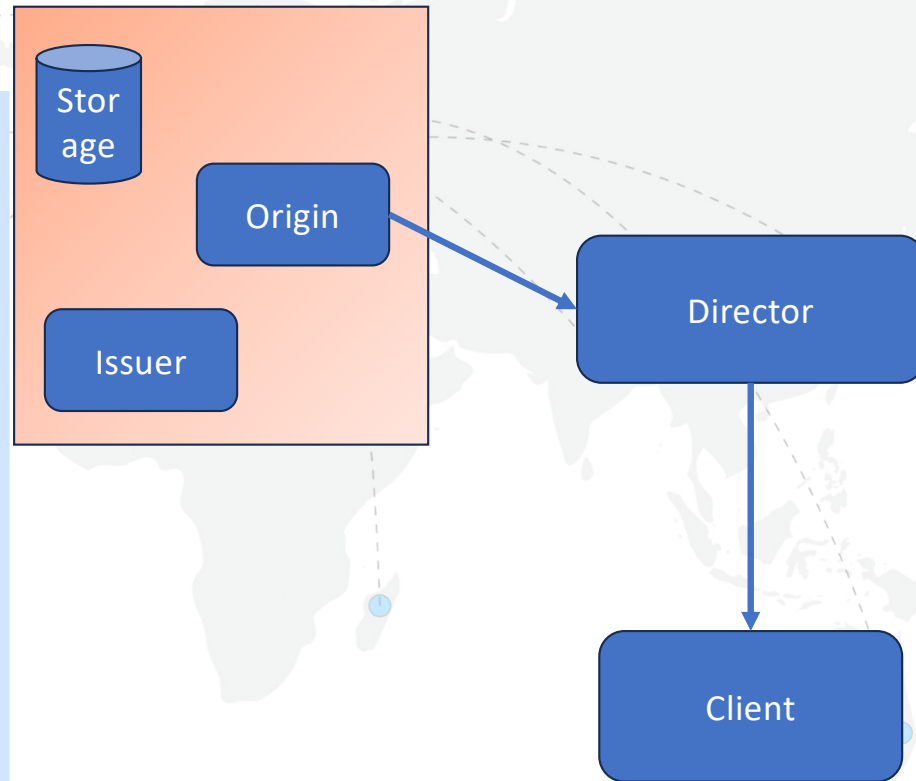


- ## What happened?



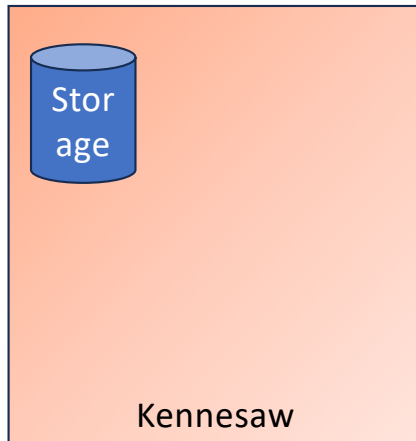
Namespace authorization

- Each origin in the OSDF can select a required authorization service for their namespace.
 - To gain access, you must have a credential from this service!
 - The origin tells the director the selected service so the director can inform clients.
 - Clients can automatically bootstrap the credential from this information.
- Issuer can be part of a larger service or utilize a built-in component in the origin.





Namespace Authorization

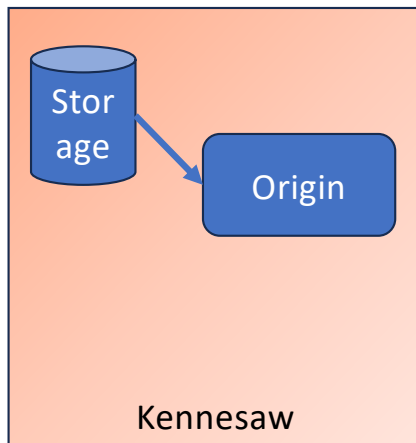


"I want my storage at Kennesaw State University available to the national community."





Namespace Authorization

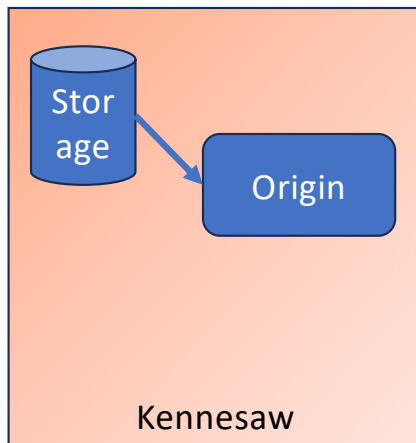


“Ok, I have setup an OSDF origin in front of your storage! Your data is available at /kennesaw in the OSDF”





Namespace Authorization

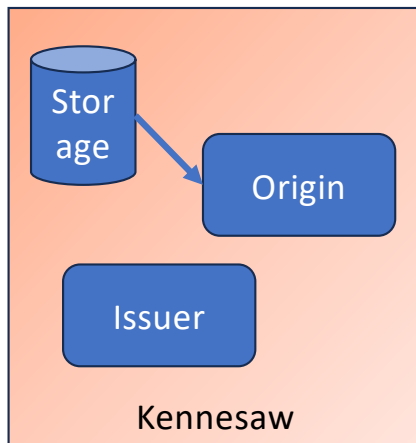


"I want my dataset at /FlamingoSim to be shared with my friend, Christina Koch."





Namespace Authorization



“Ok, I have configured the credential issuer built-in to your origin.”



“Christina, you can download the dataset from the OSDF at `/kennesaw/FlamingoSim` using the pelican client.”

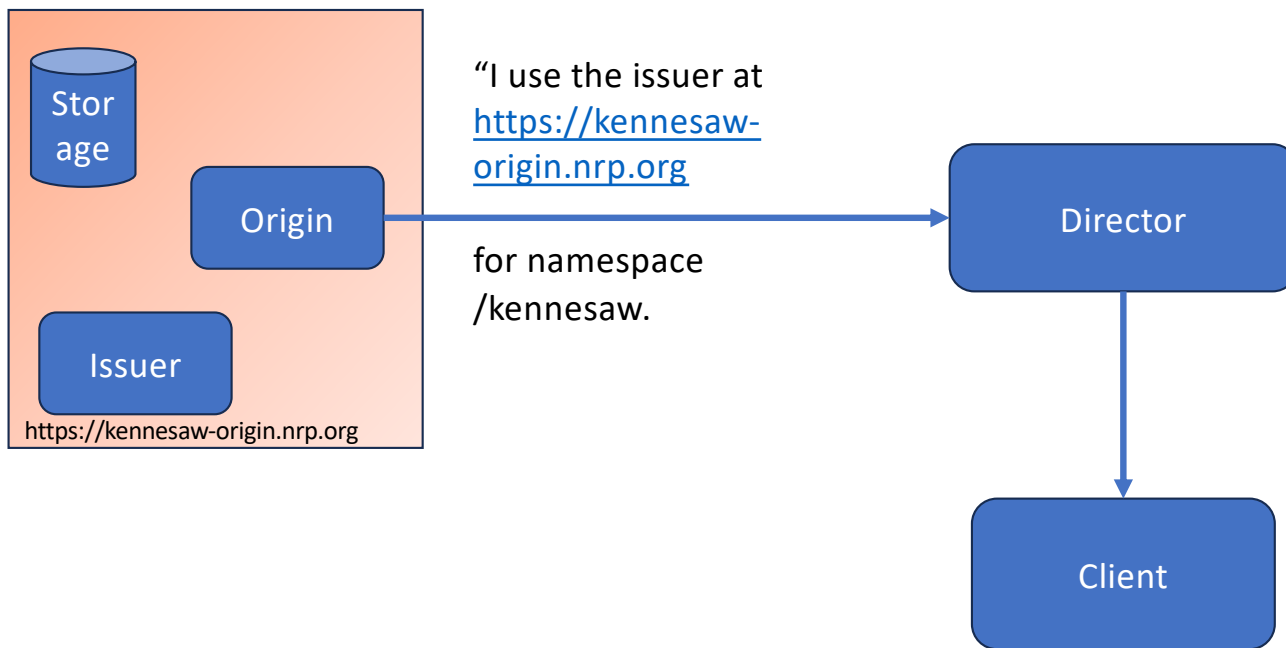


Great!



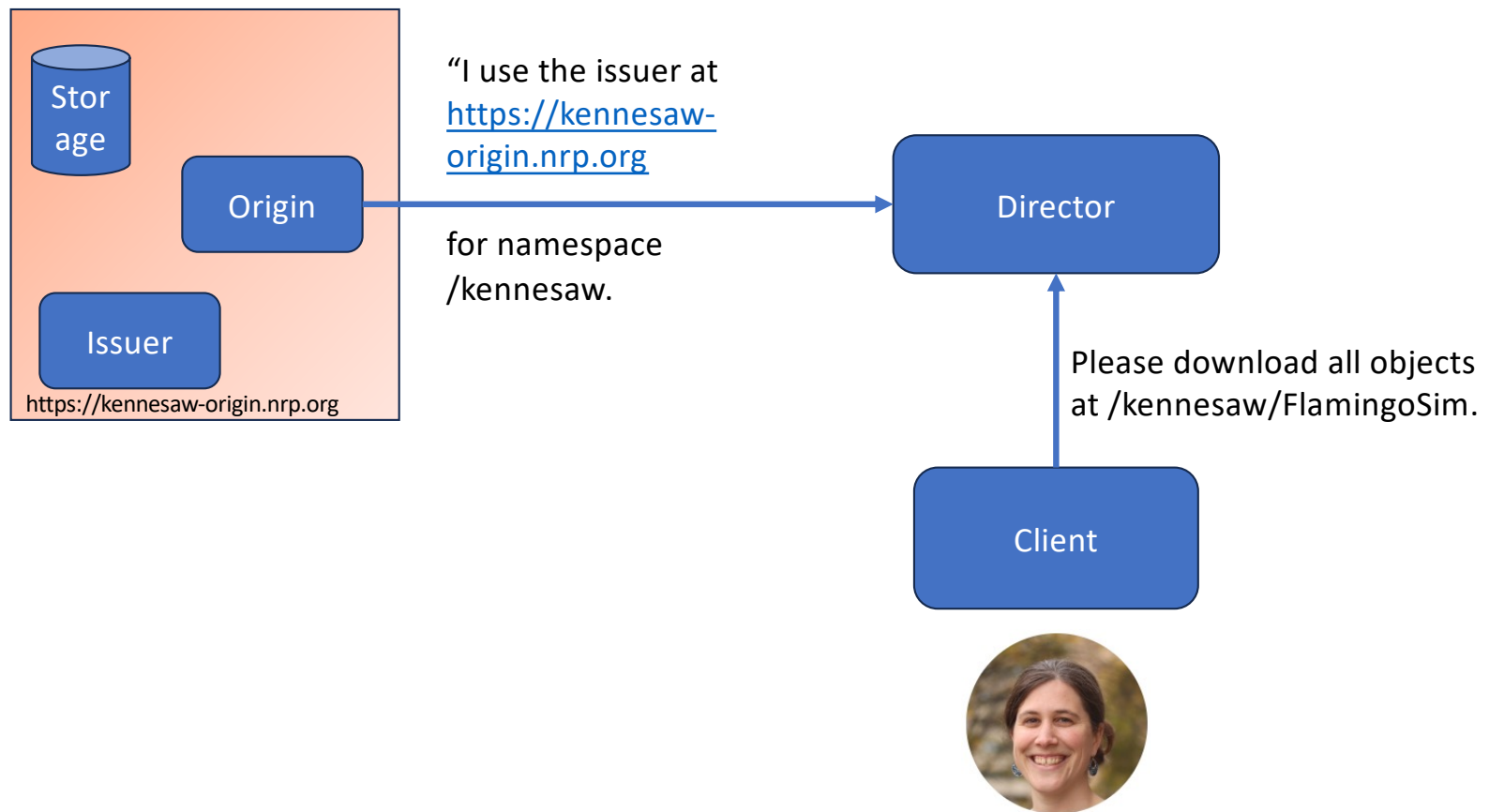


Namespace Authorization



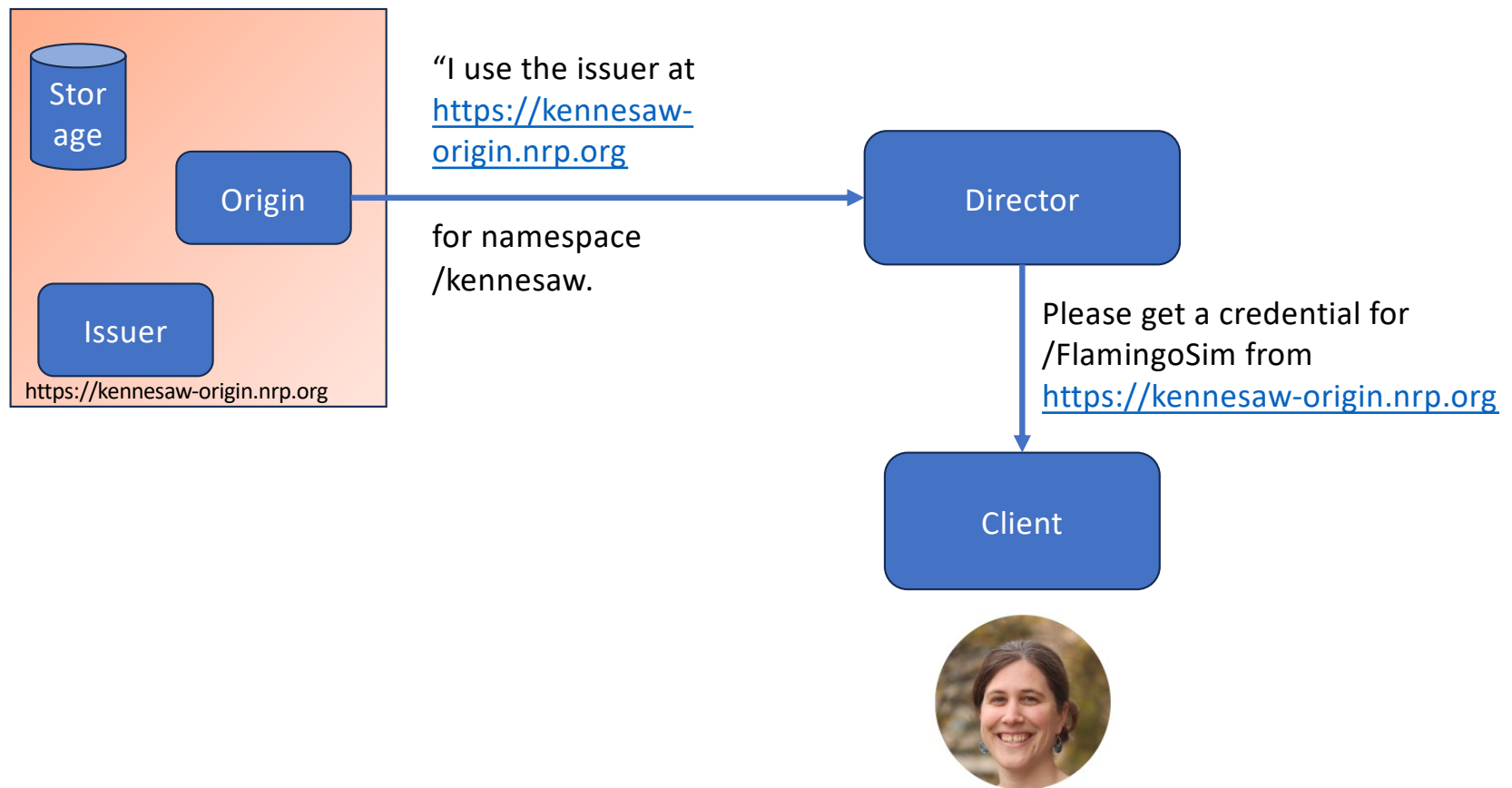


Namespace Authorization



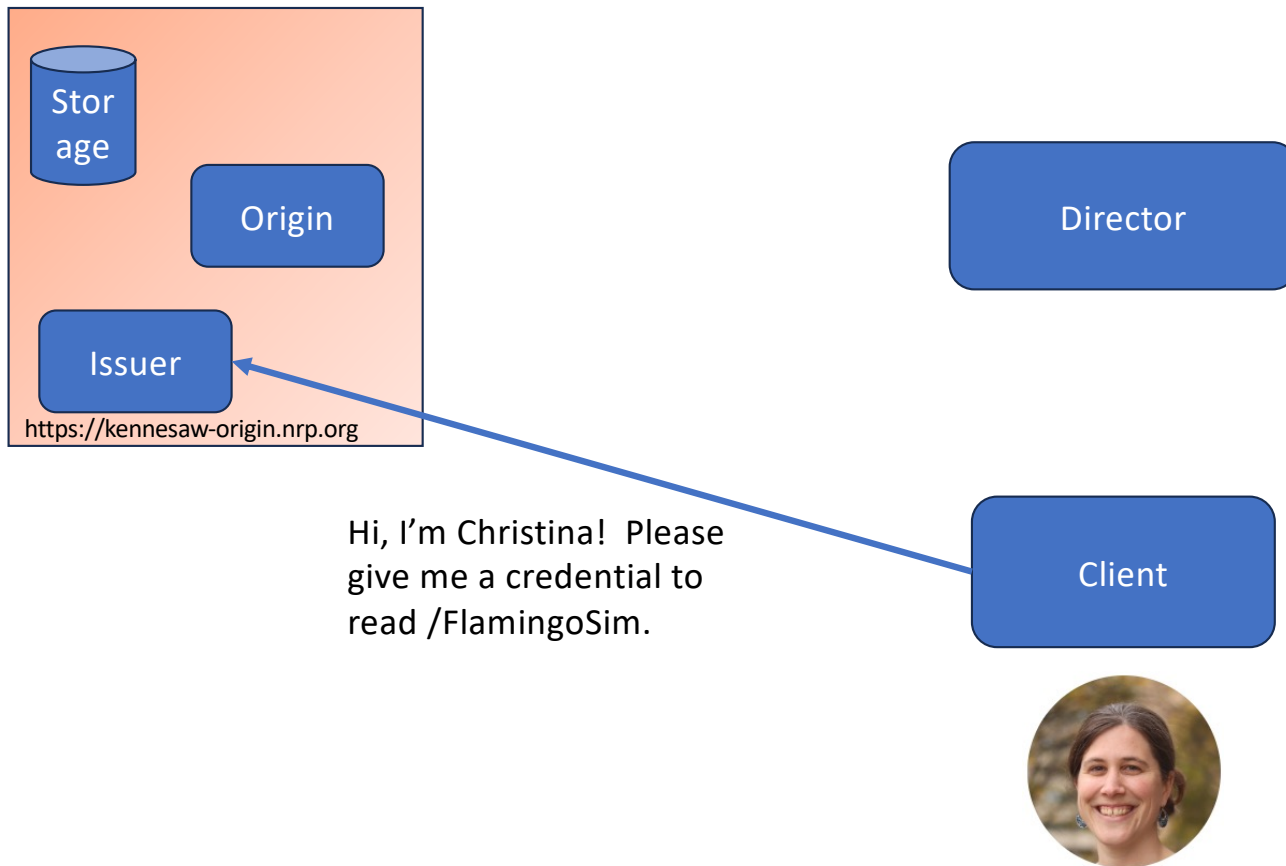


Namespace Authorization



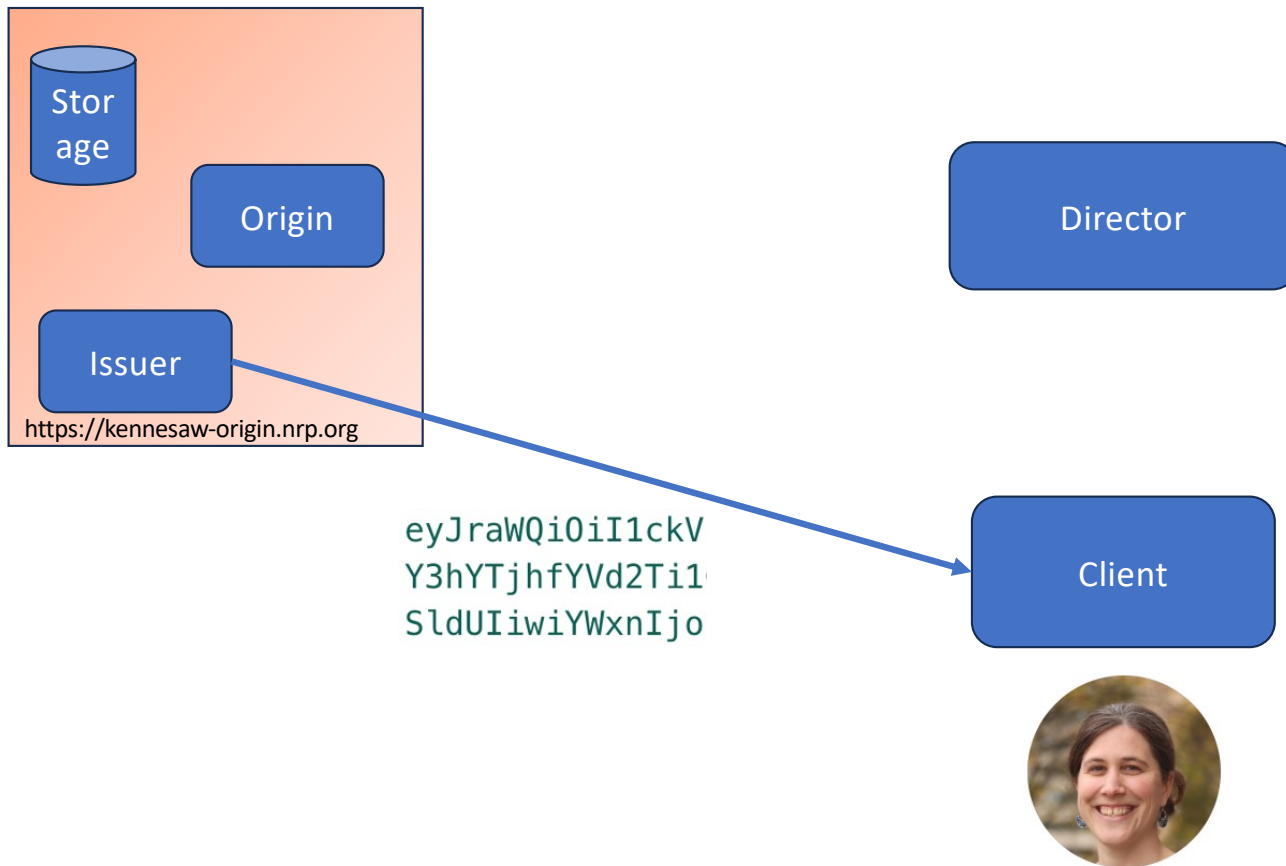


Namespace Authorization





Namespace Authorization



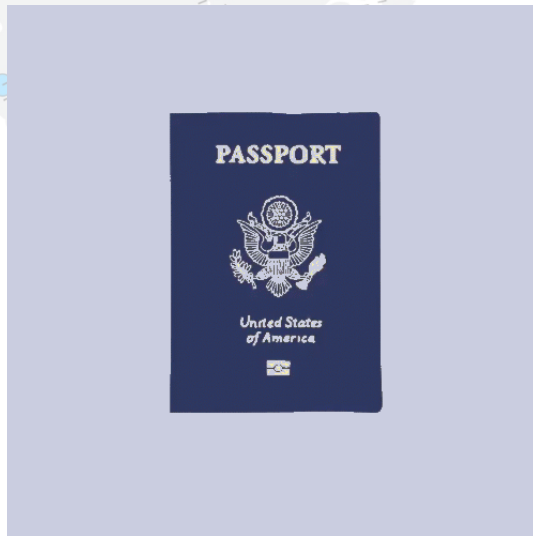


- The credentials used are in the form of a “bearer token” using a format called “JSON Web Token” (JWT).
- “Bearer token” -> bearer (client) presents the credential (token) to the server to gain authorization to the service.
- Within the OSDF, we use a **capability-based** authorization infrastructure.



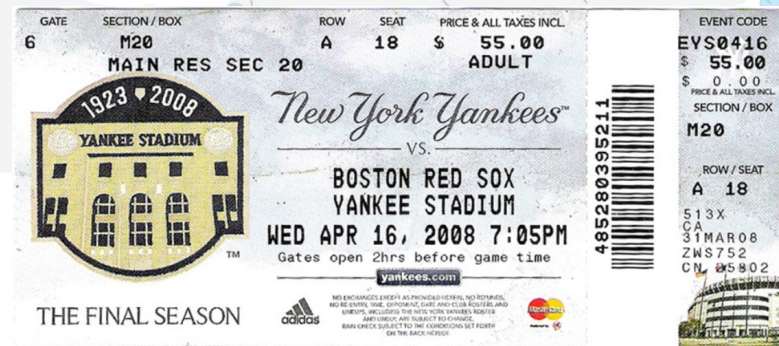
Identity versus Capabilities

The identity credential says “who you are”



The remote service uses this information to determine “what you can do”.

The capability credential says “what you can do”



The remote service determines whether (or not!) it accepts the capability.
No identity in the decision-making!



What does our token say?

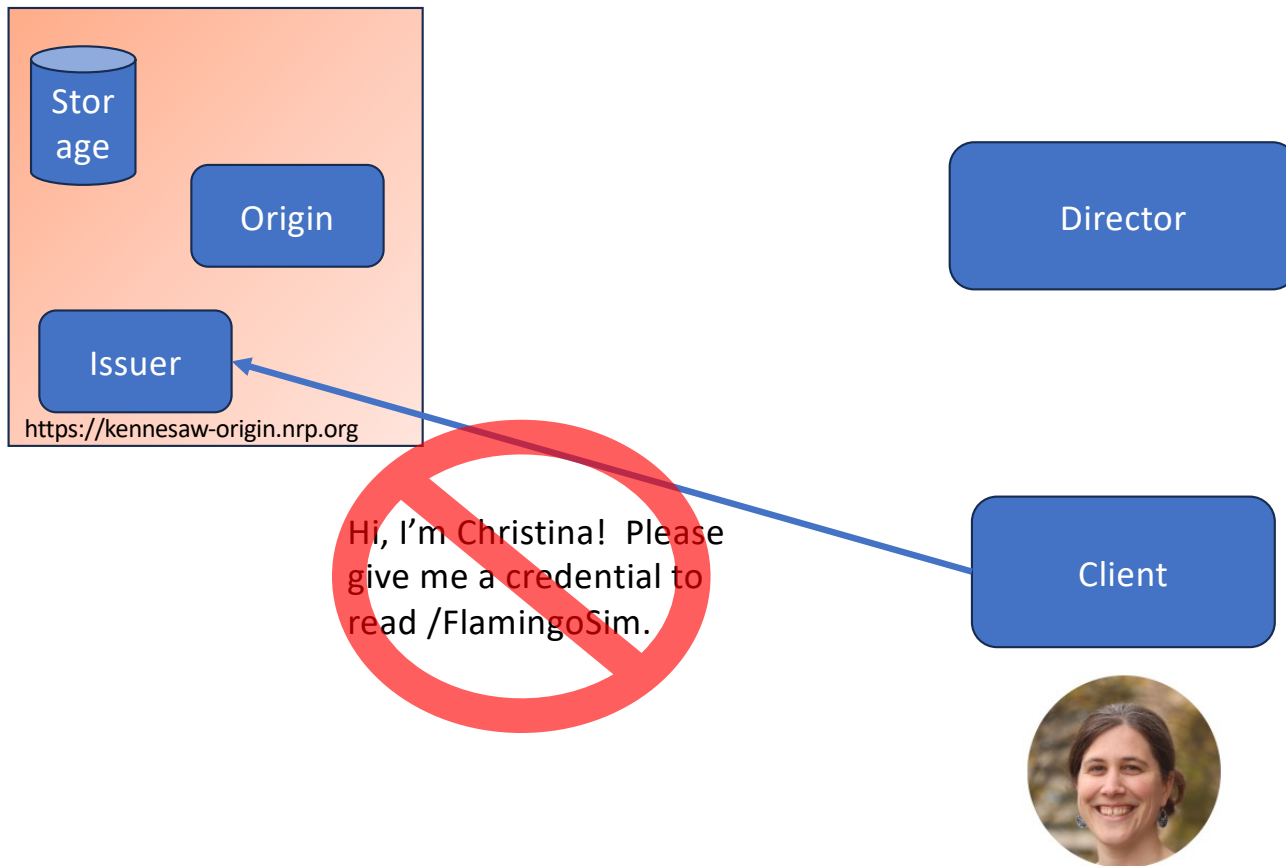
The screenshot shows the JWT Debugger interface. On the left, a long JWT token is displayed. On the right, the 'DECODED PAYLOAD' section is active, showing the token's claims in JSON format. The claims include version, audience, subject, nonce, scope, issuer, expiration, issued at, and issuer ID.

```
{
  "wlcg.ver": "1.0",
  "aud": "https://wlcg.cern.ch/jwt/v1/any",
  "sub": "http://cilogon.org/serverA/users/9265706",
  "nbf": 1752763669,
  "scope": "storage.create:/protected",
  "iss": "https://tutorial-origin.svc.sg-htc.org",
  "exp": 1752764574,
  "iat": 1752763674,
  "jti": "https://tutorial-origin.svc.sg-htc.org/api/v1.0/issuer/5372324452b7db85fe47f56d401f7b9a?type=accessToken&ts=1752763674556&version=v2.0&lifetime=900000"
}
```

- Revisit the token in the notebook at <https://jwt.io>.
- Our credential, “in plain English”, says:
 - When it expires,
 - Who should use it,
 - A unique ID, AND
 - A list of permissions (an ‘access control list’ or ACL).
- In this case, the ACL says **storage.create:/protected**

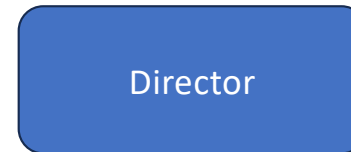
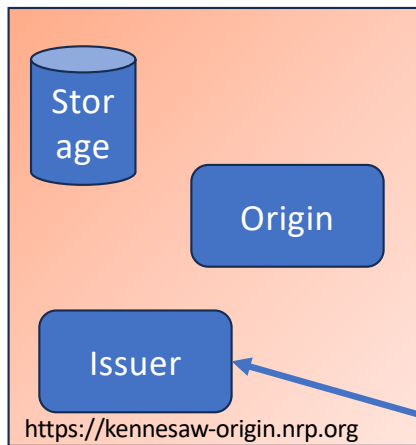


Revisiting Christina...

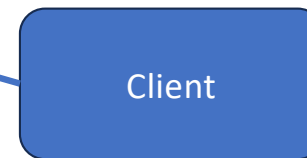




Revisiting Christina...

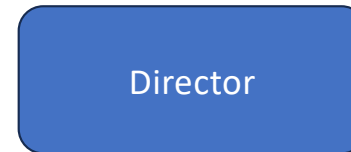
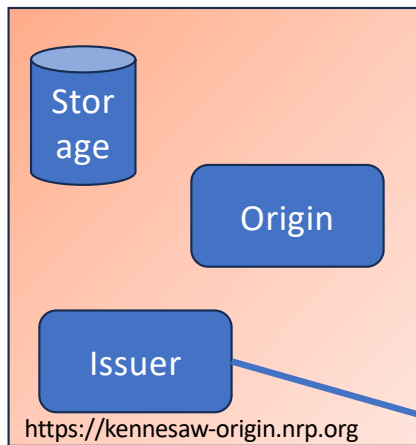


Please give me a
credential to read
/FlamingoSim.

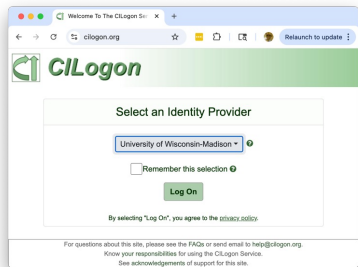
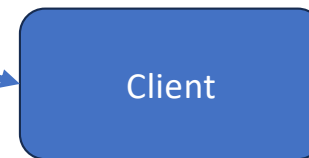




Revisiting Christina...

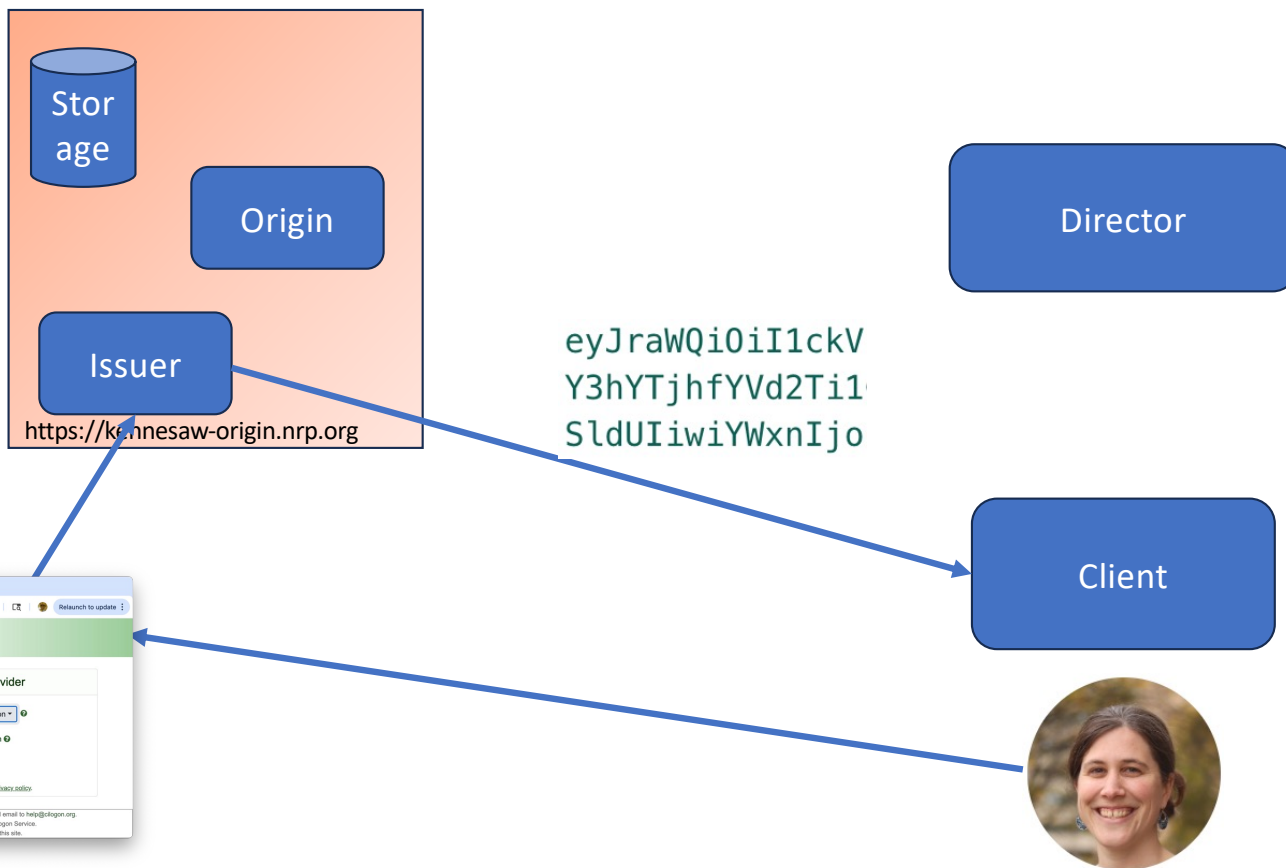


Please authenticate
at my SSO to identify
yourself





Revisiting Christina...



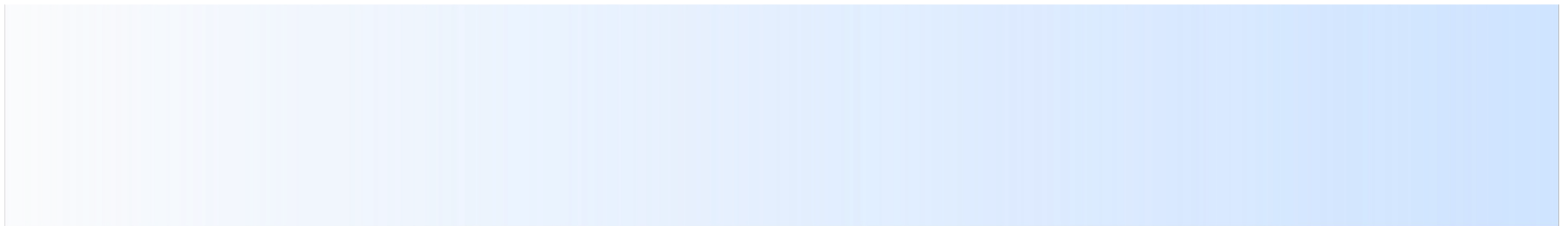


And now Christina can download the objects!





Sharing on the OSDF

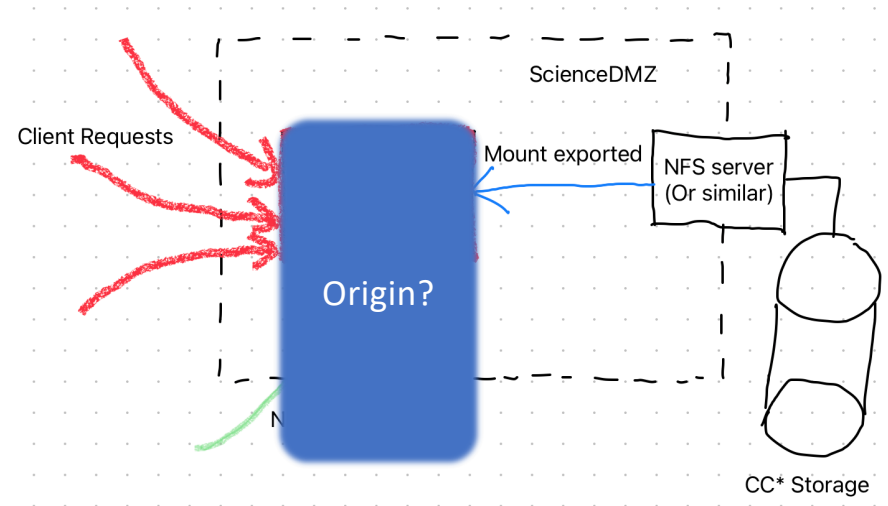




Let's say you have an Object Store...

- Suppose you have a new storage system (possibly funded by the NSF's [CC* Storage](#) program) ... what next?
 - Can be an S3 instance, POSIX mount, or Globus collection.
- The *origin* service connects the object store to the data federation.

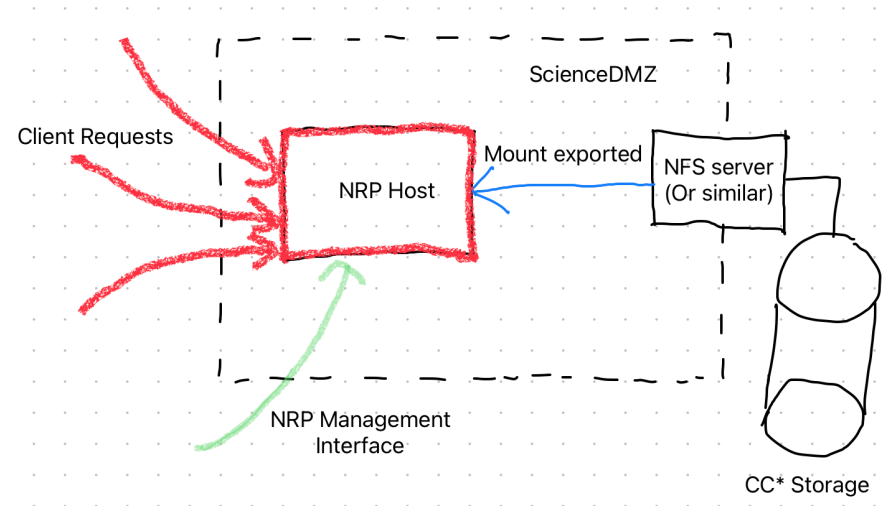
How do you get an origin?





... Then you need an origin!

- You can download the Pelican software, install it, configure it, and join it to the OSDF.
 - But then you'll need to monitor it, reboot it, upgrade it, debug it...
- OSDF partners with the National Research Platform (NRP) which runs a US-wide distributed Kubernetes cluster.
- **Sales pitch:** You provide the hardware, power, and networking; NRP will do the rest.
 - NRP provides the OSDF operators with a Kubernetes-based interface; **we do the installation and configuration.**





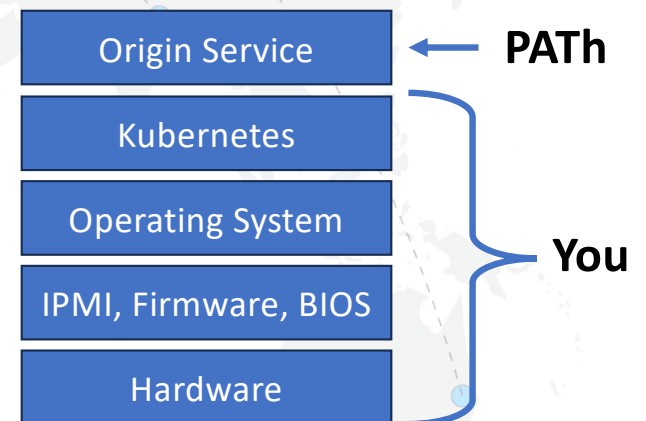
Technical Options for Kubernetes

You decide how much involvement and control you need.

Two options for Kubernetes:

1. Allocate hardware at your site and allow the NRP to manage it.
 - NRP handles everything above the hardware to Kubernetes, then PATH beyond that.
2. Operate your own Kubernetes cluster and provide
 - Onus is on your local team to manage the OS and run, upgrade, monitor the Kubernetes service.
 - A reasonable estimate is 0.25+ FTE of a skilled Linux sysadmin.
 - Often makes the most sense if you're already doing it.

“Roll Your Own” case



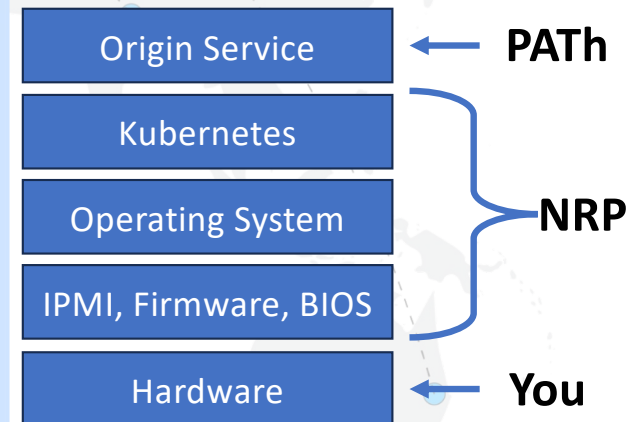
A traditional RPM-based installation is always available too!



More **NRP** Details

- If you have NRP manage the host, they go from “remote access on up”:
 - You are responsible for power and network connectivity (primary and management networks).
 - We work to arrange access to the IPMI / baseband management interface of the host for the NRP team.
 - NRP installs and manages the OS.
 - NRP installs Kubernetes and connects the host to the distributed ‘Nautilus’ Kubernetes cluster.
- Nautilus is a single cluster distributed over dozens of institutions.
 - A Cat-II Prototype system funded by the NSF.
 - Includes hardware (GPUs, FPGAs, caches) funded via the NSF but a *large* portion of hardware is brought into the collective by universities – like these integrations!

“Roll Your Own” case





Trust Relationships

Object store trusts the origin to:

- Implement its authorization policies.
- Read/write objects, potentially with privilege.
- Adhere to rate-limiting.
- Advertise the desired issuer and namespace to the federation.

Object store:

- Provides access credentials (S3) or mount point (POSIX).
- Desired policy information.
- Agrees to abide by the AUP.



Example Config Snippet

- This is a snapshot of the Kennesaw origin's configuration file.
- Specifies:
 - Hostname, port.
 - Issuer to use.
 - Enables local issuer.
 - Three namespaces to export, /kennesaw, [/noaa/fisheries-1](#), and [/routeviews](#).
- A single origin can export multiple namespaces!

```
tiger-osg-config — bbockelm@tigermaster0004:~ — vim /tmp/50-instance.y...
--
Server:
  Hostname: kennesaw-origin.nationalresearchplatform.org
  IssuerUrl: https://kennesaw-origin.nationalresearchplatform.org:8080

Origin:
  Port: 8443
  StorageType: posix
  Exports:

  # Datasets produced by researchers at Kennesaw State University
  - FederationPrefix: /kennesaw
    Capabilities: [PublicReads, Writes, Listings, DirectReads]
    StoragePrefix: /mnt/origin/kennesaw

  # Partial replica of https://www.ncei.noaa.gov/products/water-column-sonar-data
  - FederationPrefix: /noaa/fisheries-1
    Capabilities: [PublicReads, Listings, DirectReads]
    StoragePrefix: /mnt/origin/noaa/fisheries-1

  # https://support.opensciencegrid.org/a/tickets/79720
  - FederationPrefix: /routeviews
    Capabilities: [PublicReads, Listings, DirectReads]
    StoragePrefix: /mnt/origin/routeviews/data

EnableOIDC: true
EnableIssuer: true
```



Example Config Snippet

- The configuration carries the authorization information inside the namespaces.
 - List of prefixes, permitted actions, and list of identifiers (user or group names).
 - Simple templating is permitted; i.e., /home/\$USERNAME.
 - These control the issuer's policies.
- (Not shown) Controls which single sign on (SSO) is used to authenticate users.
 - With federated SSO such as Cilogon.org, this becomes a powerful way to share with collaborators.

```
tiger-osg-config — bbockelm@tigermaster0004:~ — vim /tmp/50-instance.y...
Issuer:
AuthorizationTemplates:

# https://support.opensciencegrid.org/a/tickets/79903
- prefix: /CODELab
  actions: [read, create, modify]
  users:
    - !kennesaw.edu
    - !students.kennesaw.edu

# https://support.opensciencegrid.org/a/tickets/79916
- prefix: /FlamingoSim
  actions: [read, create, modify]
  users:
    - !kennesaw.edu
    - !students.kennesaw.edu

# https://support.opensciencegrid.org/a/tickets/79916
- prefix: /iotas
  actions: [read, create, modify]
  users:
    - !kennesaw.edu
    - !students.kennesaw.edu

# https://support.opensciencegrid.org/a/tickets/80091
- prefix: /modelCproject
  actions: [read, create, modify]
  users:
```



Revisiting Origin Operations

Advantages of remotely-operated origin:

- Minimal operational load (monitoring, setup, configure).
- Minimal expertise needed (debugging).
- Larger operations team.

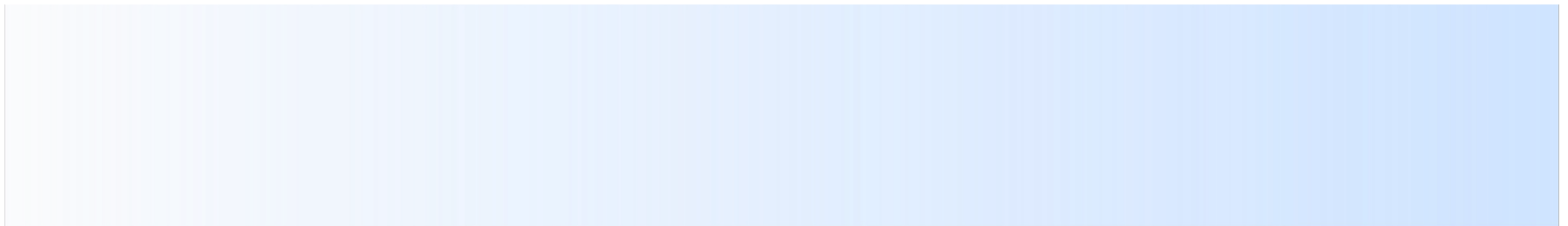
Disadvantages:

- Higher level of trust.
- Requires remotely-operated hardware (can conflict with institutional policies).
- Changes must go through the OSDF's issue tracker.

No free lunches!



The OSStore and OSDF





No place to store your data?

No problem!

- The OSDF contains minimal storage (just the caching/staging servers).
- The OSStore service provides allocations on participating object stores across the US.
- Allocations are time & space limited. E.g., “100TB for 6 months”.
- **Appropriate for:** copies of data, data that can be reproduced.
- **Not appropriate for:** archival storage, restricted data.



OSStore – a bazaar for storage

- Over the last two years, the NSF's CC* Storage program has provided funding to institutions to purchase storage services.
- **Caveat:** 20% of the space must be given to “extramural usage”.
 - This tends to be ~1PB per awardee. About half-dozen awardees currently participating.
- The OSStore helps awardees meet this requirement!
- The PATH project runs an origin for the storage, manages user allocations, and maps allocations to storage.



OSStore – What Info do you provide?

You specify:

- The PI & institution for the allocation.
- A description of the science.
- Requested storage size and duration (<2 years).
- Requested namespace prefix (/my-great-project).

Via CILogon, you manage the group of users who can read or write to the space.

We:

- Select storage location for the space.
- Operate the necessary service (origin).
- Configure the permissions (public / private).
- Setup user groups for the PI to manage.
- Provide training or computing capacity (OSPool) to use the data!



OSStore Allocation

Goal is a three-step process:

1. Request allocation, get approval, and ~wait~ for operations team to setup the origin. < 1 work-week.
2. PI is added as the administrator to two groups in CILogon; one for read-only access, one for read-write access. They add their collaborators as appropriate.
3. Start running `pelican object sync` to move the datasets into the OSStore!



Example OSStore Allocations

RouteViews dataset: a map of the Internet, as seen by participating sites. This allows operators of major Internet services to detect changes to the map in near-real time and for researchers to understand the historical evolution of the Internet.

About **75TB**.

This is a working **copy** of data stored at U. Oregon.



IceCube: The “Pass 3” physics reconstruction of the raw data recorded by the IceCube neutrino observatory at the South Pole Station.

About **600TB** once the computation is complete.

Only copy of the data; can be recomputed from raw data on tape at TACC.

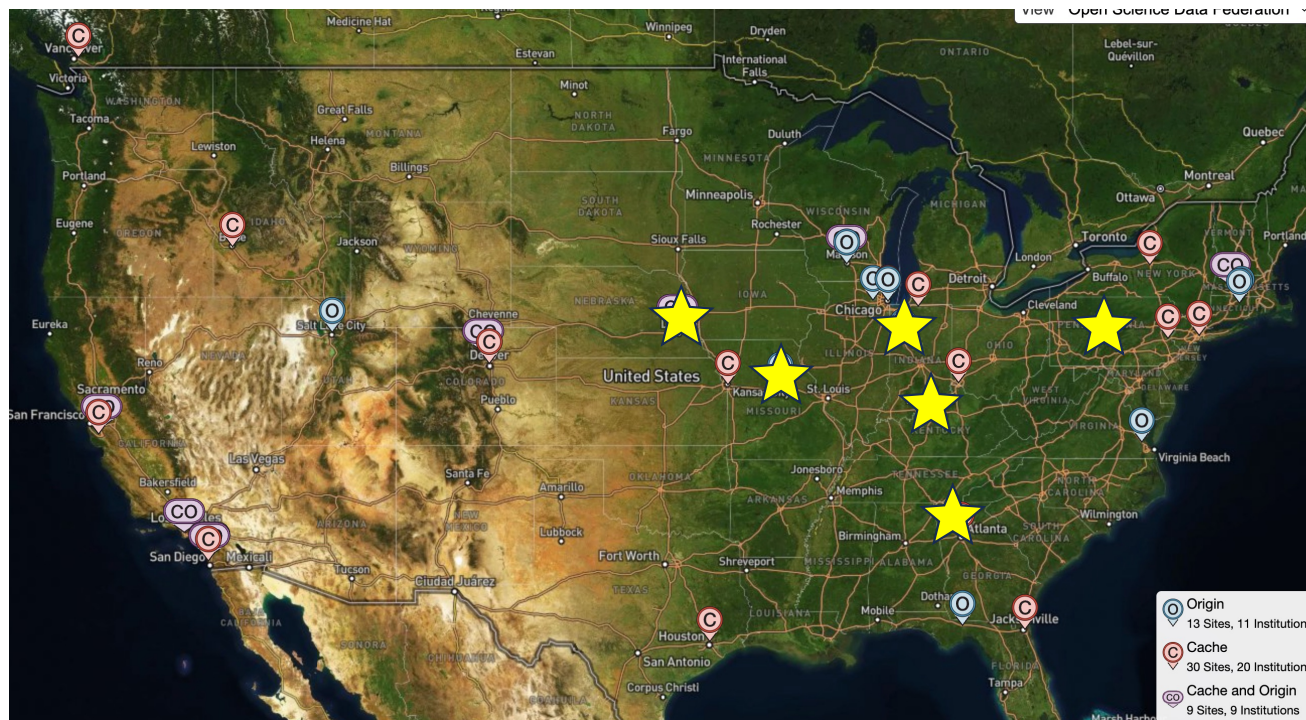


ICECUBE
NEUTRINO OBSERVATORY



Available OSStore locations

More coming online this fall!





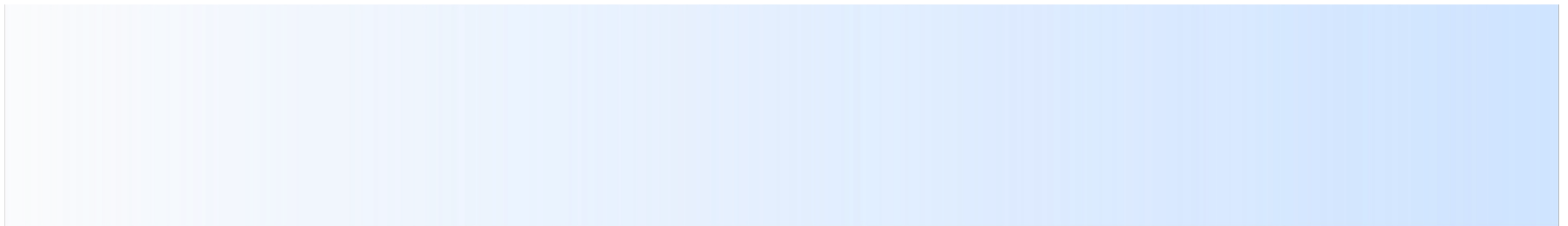
Interested in the OSStore?

- Reach out to

support@osg-htc.org



Concluding Thoughts





OSDF brings a powerful capability...

- Connecting petabytes of objects via dozens of origins.
- Accelerating delivery with dozens of caches across the US.
- Spanning major NSF & DOE investments including LIGO, IceCube, and NRAO.
- Including NCAR's Research Data Archive, a major repository of earth systems data
- We are looking to grow the user base, including you!

BYTES READ

145,932 TB

↑ 5 GB/s Last 1 Year



OBJECTS READ

3,673,507,790

↑ 116/s Last 1 Year



... powered by the Pelican Platform



The Pelican Project is turning 2! 🎉

The Pelican project started in 2023 with generous support from NSF; there are two years remaining in the current grant.

It has been a game-changer for the OSDF service. Highlights:

- Standalone: not dependent on external services to function.
- Complete rewrite of the central components.
- Origin & cache services have web management interfaces.
- Python- and Web-based clients.
- Overhaul of the client & server monitoring.
- Addition of the built-in issuer.
- End-to-end checksums.
- And lots and lots of 'polishing' out small bugs and annoying error messages!

**The outlines of the original vision are
beginning to appear!**



Many Miles to go before we rest...

Planned Highlights for the Year Ahead

- 'Collections API' – allowing the origin administrator to delegate the authorization management to a user for a sub-namespace.
 - **Goal:** Provide a more S3-like experience for managing authorization instead of config-file-based.
- Python client: Finish token discovery improvements; support writes from Python.
- Web client: Release new Javascript library & UI for non-public data browsing; built-in JupyterLab interface.
- Command-line client: “object sync” command closer to rsync in functionality.
- Simplified Origin: Remove need for X.509 server certificate and incoming network connectivity (can live behind firewalls).
- Improved Globus and S3 backends: Release overhauled Globus and S3 backends, providing improved write and object listing support.



Enjoy this tutorial?

- The OSG School is an annual, week-long event focused on how Throughput Computing can advance *your* science project.
- Throughput Computing Week is our community's annual meeting, highlighting the activities of the HTCondor Software Suite, Pelican, and OSG Consortium.





Thank you!

Questions? Comments? Thoughts?

Reach out to support@osg-htc.org.



Questions?

This project is supported by the National Science Foundation under Cooperative Agreements OAC-2331480. Any opinions, findings, conclusions or recommendations expressed in this material are those of the authors and do not necessarily reflect the views of the National Science Foundation.